

GROUP NO : 15

FINAL REVIEW

Neuro Stamp

Secure Copyright Protection Using DWT-SVD Watermarking

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Introduction [1/2]

- Neuro Stamp is a web platform designed to protect digital image copyrights.
- We use advanced mathematical frequency transformations to embed this signature and make it invisible.
- This platform empowers original creators to indisputably prove they own their work.

Introduction [2/2]

- Our core technology uses DWT, DCT and SVD to analyze the image and embed secure data into the image matrix.
- Unlike visible watermarks, our mathematical watermark cannot be cropped or easily painted over.
- This ensures copyright protection remains intact even if the image is modified, compressed, or shared across platforms.

Problem Definition [1/2]

- Social media sites like Instagram automatically erase traditional EXIF metadata from uploaded images hence can be easily copied.
- Creators rapidly lose all visible and technical proof that they originally created the content.
- Existing visible watermarks offer zero cryptographic security and are easily forged.

Problem Definition [2/2]

- A modern watermark must survive compression, JPEG conversion, scaling or added noise.
- Courts lack actionable, mathematical proof during serious copyright infringement disputes.
- We need an invisible system that survives real-world attacks and provides undeniable digital proof.

Objective of the Work [1/2]

- Implement a hybrid DWT-DCT-SVD watermarking algorithm to embed imperceptible, robust watermarks in images.
- Build a secure web application (FastAPI) with user authentication, CSRF protection, and file validation.
- Develop a copyright registry using perceptual hashing (dHash) to prevent double-registration of images.

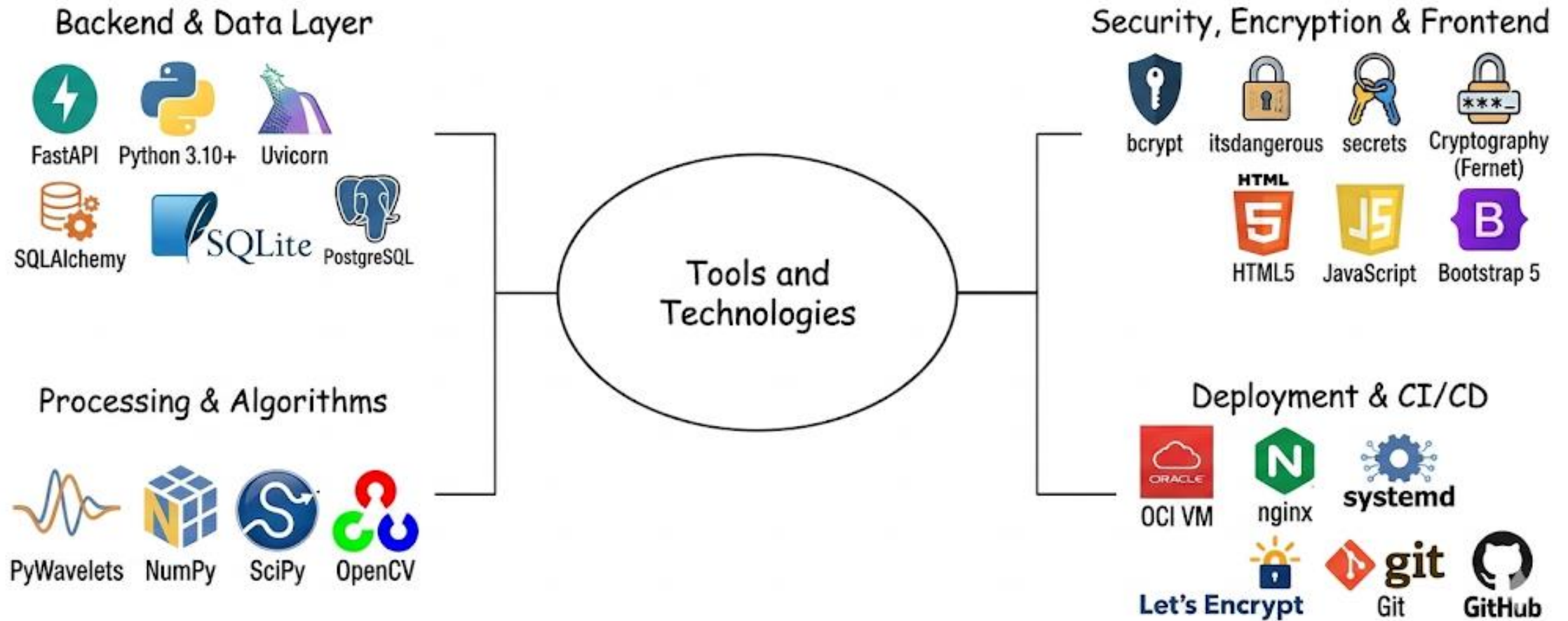
Objective of the Work [2/2]

- Create an attack simulation module to test watermark robustness against JPEG, noise, blur, rotation, crop, and scale.
- Build a visualization engine to display DWT sub-bands, SVD energy heatmaps, and difference maps.
- Deploy the application on cloud infrastructure with SSL and reverse proxy.

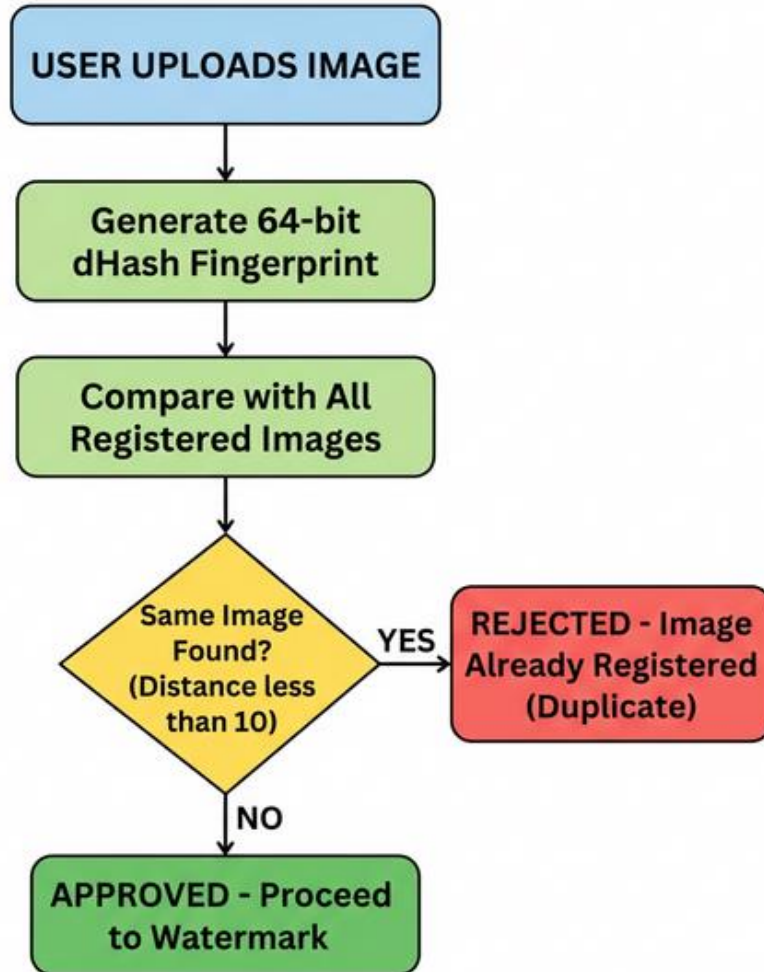
Feasibility Analysis [1/2]

- Technical Feasibility: DWT-DCT-SVD is well-researched; mature open-source libraries (PyWavelets, SciPy, NumPy, OpenCV) are available.
- Economic Feasibility: All tools are open-source (zero licensing cost).
Deployable on OCI Always-Free tier — ₹0/month.
- Operational Feasibility: FastAPI provides async I/O and built-in validation. SQLAlchemy ORM supports SQLite (dev) and PostgreSQL (prod) seamlessly.

Tool Specification [2/2]

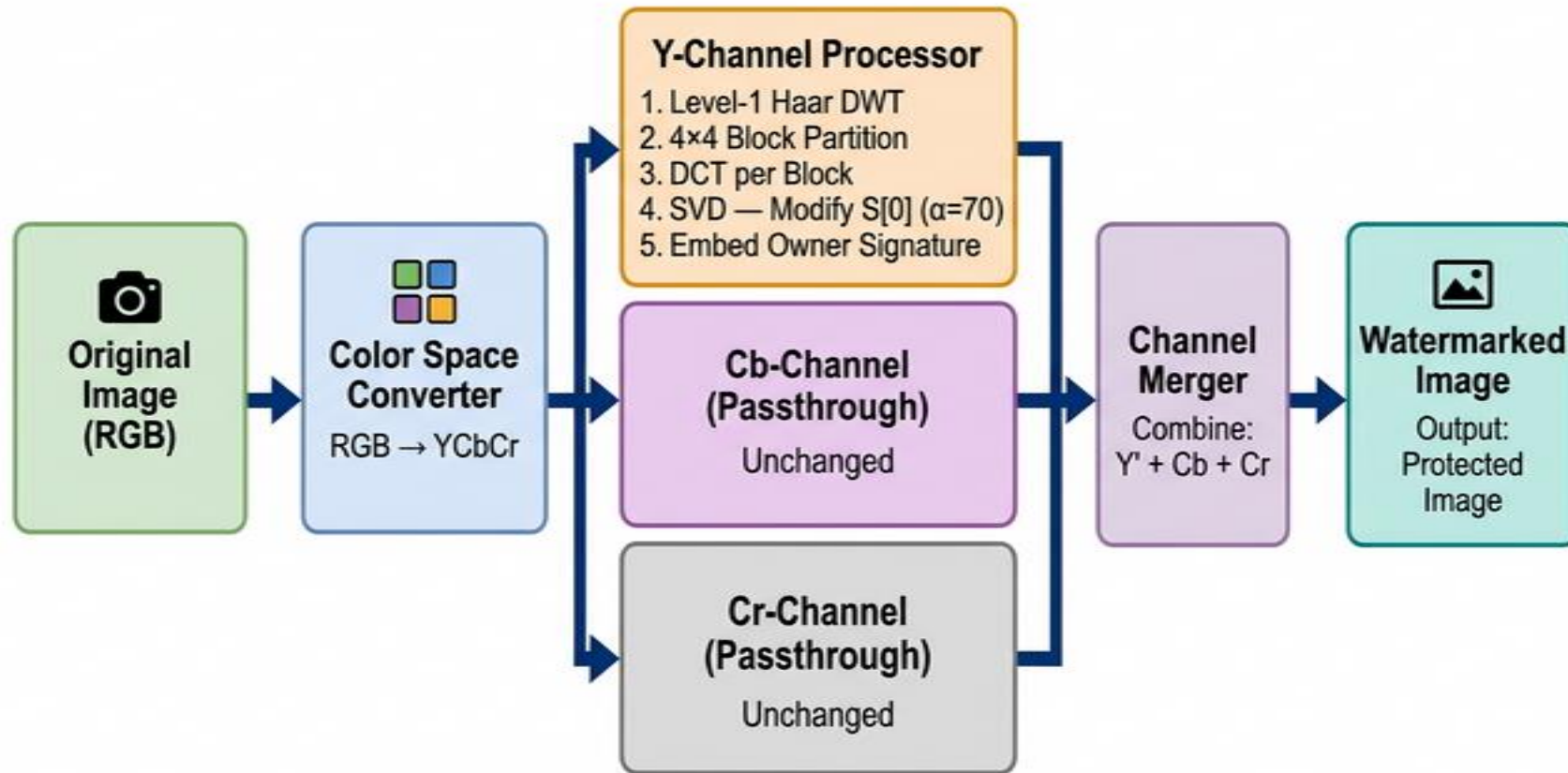


Methodology [1/4] — Authentication Module



- bcrypt password hashing (salted) + signed session tokens
- CSRF protection via double-submit cookies with timing-safe comparison
- Role-based access control via environment variables

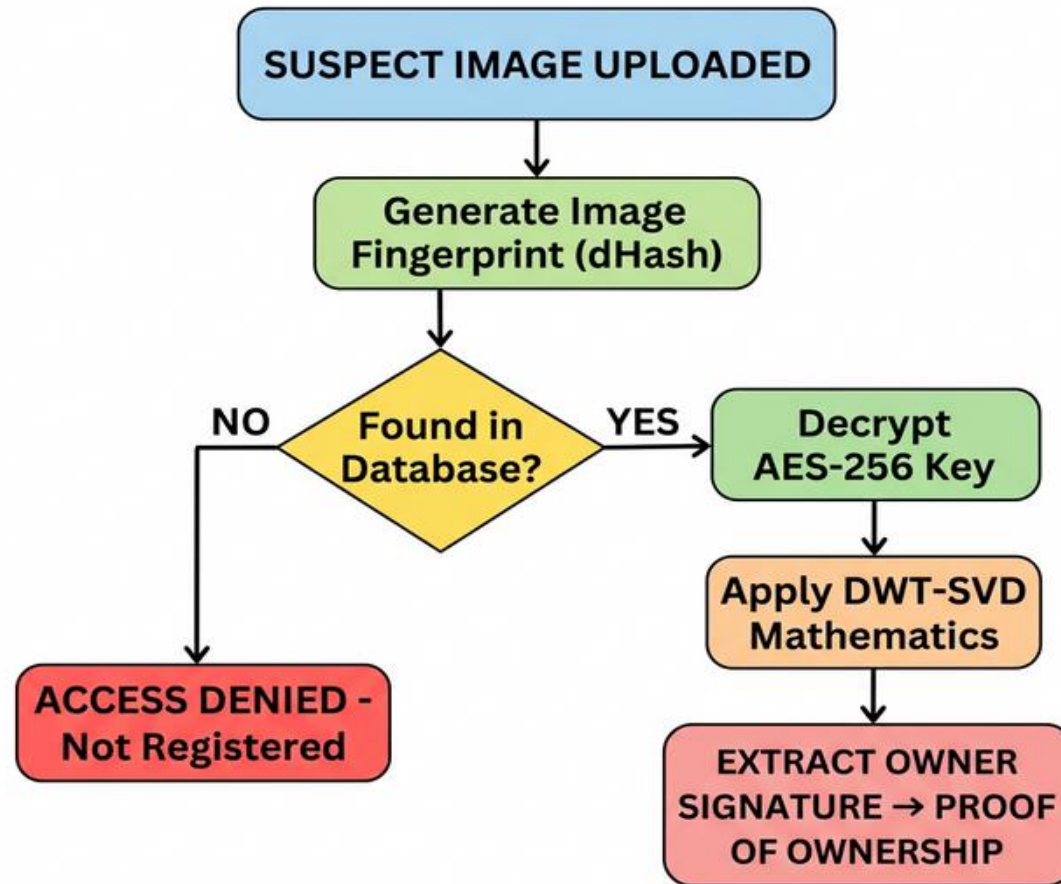
Methodology [2/4] — Watermark Embedding



- Embed: $S[0] += \alpha \times \text{bit}$ ($\alpha = 70$)

Output: watermarked image + extraction key

Methodology [3/4] — Watermark Extraction & Verification



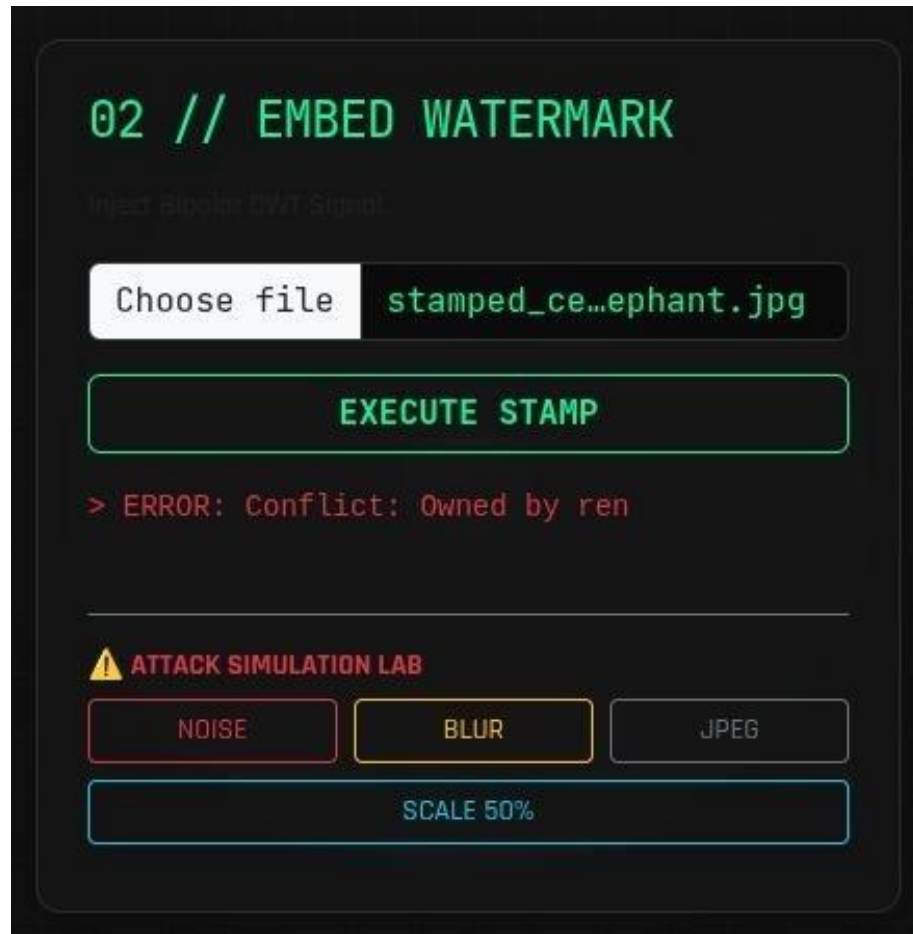
Decision rule:

if $(S[0]_{\text{current}} - S[0]_{\text{original}}) > \alpha/2 \rightarrow \text{bit} = 1$, else bit = 0

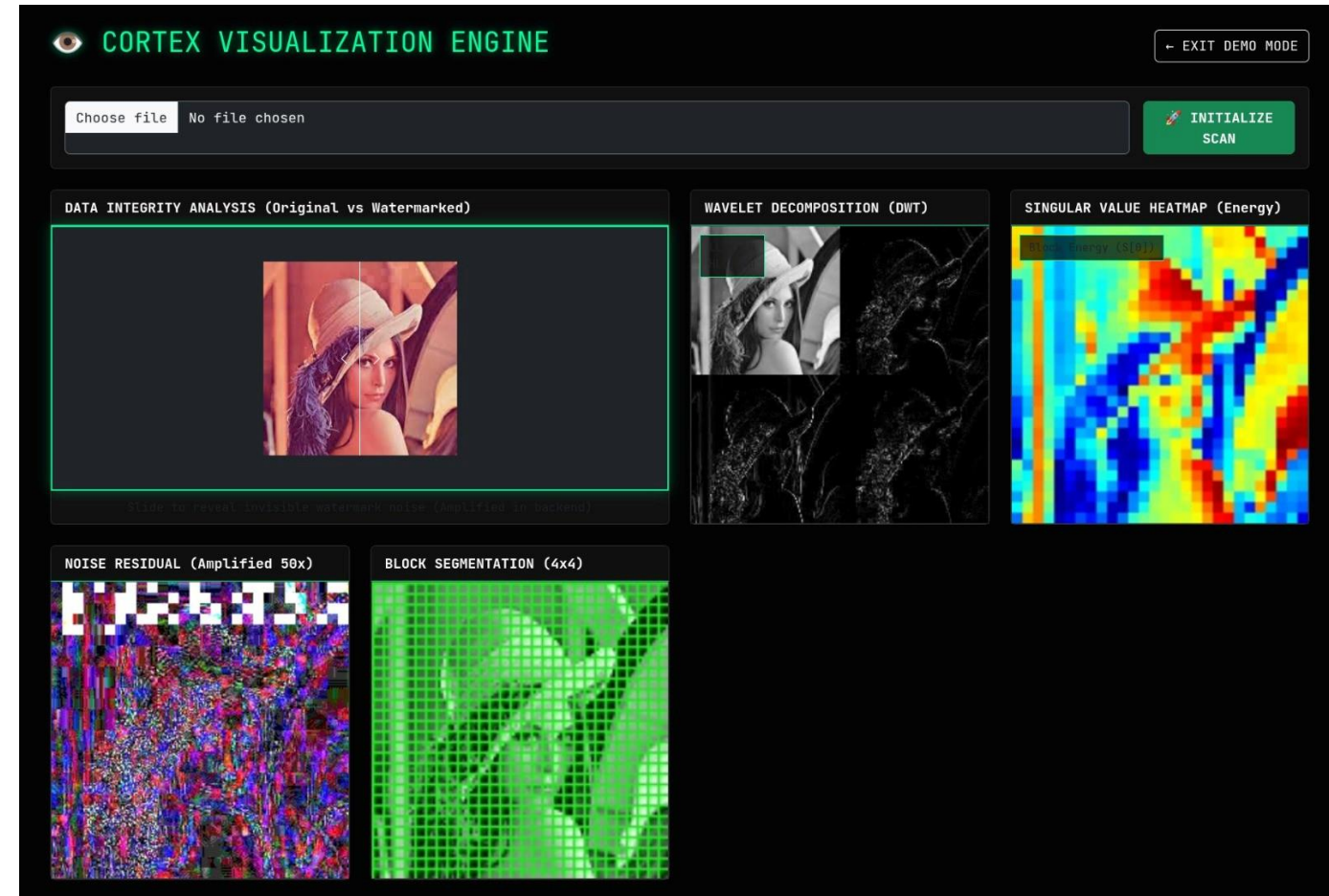
Methodology [4/4] — Visualization & Attack Simulation

- DWT sub-band decomposition views (LL, LH, HL, HH)
- SVD energy heatmap and difference map generation
- Attack simulation: Noise, Blur, JPEG, Scale (50%), Rotation

Results and Discussions [1/2]

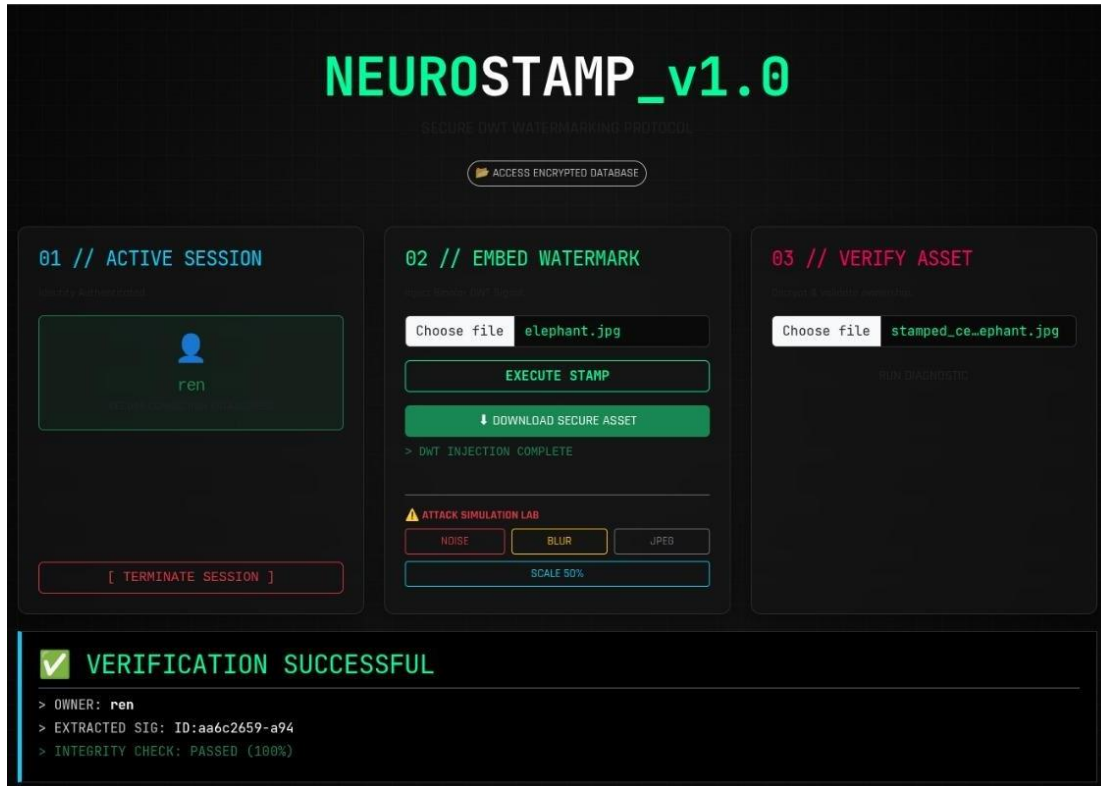


DUPLICATION DETECTION



VISUALIZATION ENGINE

Results and Discussions [1/2]



VERIFICATION ENGINE

The screenshot displays the DATABASE VAULT interface, which includes a 'VISUALIZATION ENGINE' button and a 'REFRESH' button. The interface shows two tables: 'REGISTERED USERS (Table: users)' and 'COPYRIGHT REGISTRY (Table: image_registry)'.

ID	USERNAME	PUBLIC UUID	HAS WATERMARK KEY
1	ren	aa6c2659-a94	✓ YES
2	testuser	17263dbf-9c1	✓ YES
3	testuser123	37cacc58-8c5	✗ NO
4	nithin	61e0ce14-5ed	✓ YES
5	minha	25fbc742-b8e	✓ YES

ID	PERCEPTUAL HASH (dHash)	OWNER UUID
1	898f96a4ccecac5c7	aa6c2659-a94
2	31b1afb75bc1c6c5	aa6c2659-a94
3	9f53551119793918	aa6c2659-a94
4	1be1169661119838	17263dbf-9c1
5	2e3cf09b816102a2	aa6c2659-a94
6	2f3f330707470607	aa6c2659-a94
7	824e26879b93000	61e0ce14-5ed
8	31319581c0ac8e8	61e0ce14-5ed
9	2f1f0302430e1c5b	25fbc742-b8e

DATABASE

Results and Discussions [1/2]

Attack Type	Survival Rate	Reason
Gaussian Noise ($\sigma=10$)	95%	Noise $< \alpha/2$; SVD stable
JPEG Compression (Q=50)	90%	DCT domain compatibility
Rotation $\pm 5^\circ$	85%	DWT robust to slight rotation
Gaussian Blur (r=1)	85%	Low-pass filter; SVD resilient
Scaling (50%–200%)	80%	Alignment recovery in /verify
Crop (5–10% edges)	70%	dHash matching finds owner

Error Tolerance: ≤ 32 bit errors out of 120 bits ($\sim 26\%$) $\rightarrow$ watermark still matches.

Results and Discussions [2/2]

Operation	Time
Watermark Embedding	0.4 – 1.0 sec/image
Watermark Extraction	0.3 – 0.8 sec/image
Perceptual Hash (dHash)	< 50 ms
Attack Simulation	< 500 ms
Visualization Generation	1 – 2 sec

- Watermark Payload: 120 bits (96 data + 24 CRC)
- Embedding Strength (α): 70 — balanced between invisibility and robustness
- Visual Quality: Watermarked images are visually indistinguishable from originals

SDG Mapping

SDG	Goal	Relevance to NeuroStamp
SDG 9	Industry, Innovation & Infrastructure	Applies hybrid DWT-DCT-SVD signal processing innovation
SDG 16	Peace, Justice & Strong Institutions	Protects intellectual property rights, combats piracy
SDG 4	Quality Education	Visualization engine serves as educational tool
SDG 8	Decent Work & Economic Growth	Protects digital creators' livelihoods
SDG 12	Responsible Consumption & Production	Encourages ethical use of digital media
SDG 17	Partnerships for the Goals	Built with open-source technologies

Task Distribution

Member	Primary Responsibility
Gautam G Krishna	Visualization, attack simulation, deployment infra
Goutham V E	Web app & API — FastAPI backend, AES-256, bcrypt, CSRF
Reneto Noble	Core DWT-DCT-SVD watermarking engine — embed & extract
Prajith S	Security & database No SQL, sessions, frontend UI

Conclusion

- Successfully implemented a complete, digital watermarking platform using hybrid DWT-DCT-SVD algorithm.
- Achieved 70–95% robustness across 6 attack types (noise, JPEG, blur, rotation, crop, scale).
- Built a secure web platform with 7 security mechanisms — production-hardened and deployment-ready.
- Created a perceptual hashing-based copyright registry with anti-duplication via dHash
- Developed an interactive visualization engine for algorithm transparency and education.

References

- [1] Y. Dong, R. Yan, and C. Yin, "An adaptive robust watermarking scheme based on chaotic mapping," *Sci. Rep.*, vol. 14, art. no. 24735, 2024, doi: 10.1038/s41598-024-76101-w.
- [2] H.-L. Li, X.-Q. Zhang, Z.-H. Wang, Z.-M. Lu, and J.-L. Cui, "Resampling-Detection-Network-Based Robust Image Watermarking against Scaling and Cutting," *Sensors*, vol. 23, no. 19, p. 8195, Sep. 2023, doi: 10.3390/s23198195.